

Kartikey Pathak

I build machines end to end — mechanical, PCB, firmware, control, and the app on top.

+91 7506048765 | kartikeypathak.imp@gmail.com | spongefai.vercel.app | github.com/NoobMaster-version | [LinkedIn](#)

EDUCATION

Veermata Jijabai Technological Institute (VJTI)

B.Tech, Electrical Engineering — Minor in Robotics (CGPA 8.04)

2023 – 2027

Mumbai, India

EXPERIENCE

Eyecandy Robotics

Nov 2025 – Present

Electronics / Hardware Engineer

- Design the PCB stack for a legged companion robot: compute boards (Raspberry Pi Zero 2W, Milk-V SoC), 1S/2S power and battery-charging boards (BQ25887, MP2672), a servo driver board, a mic board and inter-board connectors — schematic through layout, DFM and fabrication output.
- Integrated IMU, buffered servo bus, per-leg force-sensor conditioning, audio amp and mic array on-board, plus test points and jumper options for bring-up. Also wrote the tooling that provisions one runtime image onto a whole fleet, each robot with its own identity.

NishCorp Technologies

May – Nov 2025

Robotics Control Systems Intern

Hybrid

- Turned the CAD of a stair-climbing wheelchair into a working simulation — CAD to URDF, Gazebo/RViz, differential-drive kinematics, IMU plugin, teleop — so its motion could be studied before it was built in steel. This formed part of the case that helped the startup raise Rs 13 lakh in grants.
- Built the half-scale hardware prototype through 3D printing, assembly, motor electronics and field testing; it could sit and stand under its own power.

Krishna Defence and Allied Industries Ltd.

May – Jul 2025

Research Intern, Defence Technology

at IIT Gandhinagar

- Solved indoor localization for an autonomous forklift in a naval dockyard, where GPS is unusable and wheel odometry drifts. Chose UWB over Bluetooth, LoRa, RFID, radar and vision, then wrote custom double-sided two-way ranging firmware (Qorvo DWM3001CDK) — tuning channel, preamble, PAC size and timing delays to take usable range from the stock 45 m to **315 m**.
- Ran SLAM and NAV2 on a Raspberry Pi 4B fusing UWB, IMU and LIDAR; scaled testing from a 2×2 m room to a 50×60 m yard and validated against a surveyor's total station: **0.4% error over 10,000 m²**.

PROJECTS

PCB Axial-Flux BLDC Motor | KiCAD, FEMM, Python, STM32 — final year project

2026 – Present

- Designing a coreless motor whose stator is the PCB: 100 mm 2-layer board, 6 programmatically generated wedge coils over 8 poles, dual 3D-printed rotor, 16 N42 magnets, 0.5 mm air gap.
- Built the magnetic model myself: a top-down 2D cut cannot yield torque, so I unrolled it into radial slices to put coil current out of plane, then pulled Kt, Kv and ripple from the air-gap flux. It caught a design error pre-fabrication — the plastic rotor gives away $3.3\times$ the torque of the steel-backed design it was sized for.
- Driver firmware on an STM32G431 + DRV8311H: 20 kHz 3-phase PWM, 40 kHz control ISR, AS5600 encoder over I2C, DMA phase current sensing.

Smart Switch Board | ESP32, ESP-IDF, FreeRTOS, MQTT/TLS, BLE, Flutter, CAD

2025 – Present

- Built an internet-controlled mains wall switch across every layer alone — CAD enclosure, a relay and optocoupler board isolating 220 V from a 3.3 V MCU, firmware, a TLS MQTT broker, a Flutter app — under a hard rule that the physical switches keep working with WiFi, cloud and app down. Installed and running at home.

e-Yantra Warehouse Drone, IIT Bombay | ROS2, OpenCV, ArUco, PID

Oct 2024 – Jan 2025

- Tuned PID control for stable hover and precise motion, and wrote the vision that decodes ArUco markers for localization and pick-and-place in a simulated warehouse. **National semifinalist.**

MARIO — 3-DOF Teaching Arm | ROS2, Gazebo Fortress, micro-ROS, ESP32, C++

Jul – Aug 2024

- Ported my robotics society's teaching arm off end-of-life Gazebo Classic, rewrote teleop for beginners, and unified the ESP32/micro-ROS pipeline so one command set drives both sim and hardware. Still runs the ROS2 workshops.

Also built | vision/ML, embedded, mechanical

2023 – 2025

- **Coin sorter** (24 h hackathon — CNN on a Pi routes coins by servo gate, above 95% on Rs 5 and Rs 10), **voice-controlled pick-and-place arm** (speech to detection to inverse kinematics), **Kurma** quadruped (statically stable turtle gait), **self-balancing robot**, **maze solver** (4th), **gesture-controlled car** (glove IMU over ESP-NOW), a cat feeder and a river-cleaning boat.

LEADERSHIP & TEACHING

- **Society of Robotics and Automation, VJTI** — Active Member, Aug 2024 – Mar 2025. Taught workshops on embedded C, ESP32 and sensor fusion and mentored juniors through their first robots; the society builds the kits these run on, and **over 200 students** have learned on them.
- **Pratibimb, VJTI cultural festival** — Department Head, Electrical Engineering, Aug 2023 – Mar 2025. Led the branch to its first overall win in the fest's 26-year history. Also directed a short film, *Plot Se Panga*, which won first prize across all branches.

TECHNICAL SKILLS

Languages: C, C++, Python, Dart, Lua, TypeScript

Robotics, control & embedded: ROS/ROS2, Gazebo, RViz, URDF, micro-ROS, SLAM, NAV2, PID, sensor fusion, kinematics, FOC; ESP32 (ESP-IDF, FreeRTOS, ESP-NOW), STM32, Raspberry Pi, I2C/SPI/UART, PWM, ADC/DMA, BLE, MQTT

Electronics, simulation & tools: KiCAD schematic & layout, DFM, buck regulators, battery charging, motor drivers, mains isolation, scope and logic-analyzer bring-up; FEMM magnetostatics, OnShape, 3D printing, OpenCV, ArUco, TensorFlow, Linux, Git, Astro